

Example 4 : (a) Express $|3x-2|$ without using the absolute value symbol

(b) $h(x) = |3x^2 - 2x|$

$$f(x) = \left| \frac{x+1}{2-x} \right|, x \neq 2$$

Theorem :

Suppose $a > 0$ then

- (i) $|x| = a$ if and only if $x = \pm a$
- (ii) $|x| < a$ if and only if $-a < x < a$
- (iii) $|x| > a$ if and only if $x > a$ or $x < -a$

Example 5 : Solve

(a) $|5x-2| = 3$; (e) $|x-1| < -3$

(b) $|x+3| \geq 4$; (f) $|x^2-9| > 14$

(c) $|2x-7| < 8$; (g) $\frac{x}{|x+1|} \geq 0$

(d) $|3x-2| > -5$; (h) $\left| \frac{x+1}{x-3} \right| > 0$

Example 4 :

$$(a) |3x-2| = \begin{cases} 3x-2, & \text{if } 3x-2 \geq 0 \\ 2-3x, & \text{if } 3x-2 < 0 \end{cases}$$
$$= \begin{cases} 3x-2, & \text{if } x \geq \frac{2}{3} \\ 2-3x, & \text{if } x < \frac{2}{3} \end{cases}$$

(b) "Will not be discussed in class" //

$$h(x) = |3x^2 - 2x| = |(3x-2)(x)| = |3x-2| \cdot |x|$$

$-x$		x		x
$2-3x$		$2-3x$		$3x-2$
	0		$\frac{2}{3}$	

$$h(x) = \begin{cases} 3x^2 - 2x, & \text{if } x < 0 \\ 2x - 3x^2, & \text{if } 0 \leq x < \frac{2}{3} \\ 3x^2 - 2x, & \text{if } x \geq \frac{2}{3} \end{cases}$$
$$= \begin{cases} 3x^2 - 2x, & \text{if } x \in (-\infty, 0) \cup [\frac{2}{3}, +\infty) \\ 2x - 3x^2, & \text{if } x \in [0, \frac{2}{3}) \end{cases}$$

$$f(x) = \frac{|x+1|}{|2-x|}$$

$$\begin{array}{ccccccc} -x-1 & & +x+1 & & x+1 & & \\ \hline x-2 & -1 & x-2 & \otimes & 2 & 2-x & \end{array}$$

$$f(x) = \begin{cases} \frac{-x-1}{x-2}, & \text{if } x < -1 \\ \frac{+x+1}{x-2}, & \text{if } -1 \leq x < 2 \\ \frac{x+1}{2-x}, & \text{if } x > 2 \end{cases}$$

$$= \begin{cases} \frac{x+1}{2-x}, & \text{if } x \in (-\infty, -1) \cup (2, +\infty) \\ \frac{x+1}{x-2}, & \text{if } x \in [-1, 2) \end{cases}$$

Example 5 :

$$(a) |5x-2|=3 \Rightarrow 5x-2=3 \text{ or } 5x-2=-3$$

$$\Rightarrow x = \frac{3+2}{5} \text{ or } x = \frac{-3+2}{5}$$

$$\Rightarrow x = 1 \text{ or } x = -\frac{1}{5}$$

$$\Rightarrow x \in \left\{-\frac{1}{5}, 1\right\}$$

$$(b) |x+3| \geq 4 \Rightarrow x+3 \geq 4 \text{ or } x+3 \leq -4$$

$$\Rightarrow x \geq 4-3 \text{ or } x \leq -4-3$$

$$\Rightarrow x \geq 1 \text{ or } x \leq -7$$

$$\Rightarrow x \in (-\infty, -7] \cup [1, \infty)$$

$$(c) |2x-7| < 8 \Rightarrow -8 < 2x-7 < 8$$

$$\Rightarrow -8+7 < 2x < 8+7$$

$$\Rightarrow -1 < 2x < 15$$

$$\Rightarrow -\frac{1}{2} < x < \frac{15}{2}$$

$$\Rightarrow x \in \left(-\frac{1}{2}, \frac{15}{2}\right)$$

$$(d) |3x-2| > -5 \Rightarrow x \in \mathbb{R}$$

$$(e) |x-1| < -3 \Rightarrow \text{No solution}$$

$$(f) |x^2-9| > 16$$

$$\Rightarrow x^2-9 > 16 \text{ or } x^2-9 < -16$$

$$\Rightarrow x^2 > 16+9 \text{ or } x^2 < -16+9$$

$$\Rightarrow x^2 > 25 \text{ or } x^2 < -7$$

$$\Rightarrow \sqrt{x^2} > \sqrt{25} \quad \text{or (no solution) } \emptyset$$

$$\Rightarrow |x| > 5 \quad (\text{or } \emptyset)$$

$$\Rightarrow x > 5 \text{ or } x < -5 \quad (\text{or } \emptyset)$$

$$\Rightarrow x \in (-\infty, -5) \cup (5, +\infty)$$

$$(g) \frac{x}{|x+1|} \geq 0$$

$$\Rightarrow x \geq 0 \text{ and } |x+1| \neq 0 \text{ since } |x+1| \geq 0$$

$$\Rightarrow x \geq 0 \text{ and } x \neq -1$$

$$\Rightarrow x \in [0, +\infty)$$

$$(h) \left| \frac{x+1}{x-3} \right| > 0 \Rightarrow x \neq 3 \text{ and } x \neq -1$$

$$\Rightarrow x \in (-\infty, -1) \cup (-1, 3) \cup (3, +\infty)$$

NB: If $a \leq 0$, the theorem should not be used. (see Eg 5: (d), (e), (g), (h))

A theorem is used if and only if all its conditions are satisfied.